AIF-C01 Complete Flashcard Deck (All 5 Domains)

175 flashcards covering all five AWS Certified AI Practitioner (AIF-C01) exam domains, organized for spaced repetition study.

Domain 1 - Fundamentals of AI and ML

Card #: 1
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is overfitting in machine learning?
Back: Overfitting occurs when a model learns the training data too closely, including its noise and outliers. The

Card #: 2
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is underfitting in machine learning?
Back: Underfitting occurs when a model is too simple to capture the underlying patterns in the data. The model

Card #: 3
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is supervised learning?
Back: Supervised learning trains a model using labeled data, where each input has a known correct output. The

Card #: 4
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is unsupervised learning?
Back: Unsupervised learning trains a model using unlabeled data, with no predefined correct answers. The mode

Card #: 5
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is reinforcement learning?
Back: Reinforcement learning trains an agent to make a sequence of decisions by rewarding desired behaviors and

Card #: 6
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is a neural network?
Back: A neural network is a computing system loosely inspired by the human brain, made up of layers of interco

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Card #: 7
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is a feature in machine learning?
Back: A feature is an individual measurable input variable used by a model to make predictions, such as age, inc

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Card #: 8
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Beginner
Front: What is a label in machine learning?
Back: A label is the known correct answer or output associated with a piece of training data. Labels are required

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Card #: 9
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Intermediate
Front: What is inference in the context of AI/ML?
Back: Inference is the process of using a trained model to generate a prediction or output on new data. It happe

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Card #: 10
Domain: 1 - Fundamentals of AI and ML
Category: AI Term
Difficulty: Intermediate
Front: What is data drift?
Back: Data drift occurs when the statistical properties of incoming data change over time compared to the data

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Card #: 11
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What is Amazon SageMaker AI?
Back: Amazon SageMaker AI is AWS's comprehensive platform for building, training, and deploying custom mac

Card #: 12
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon SageMaker Canvas?
Back: Amazon SageMaker Canvas is a visual, no-code interface within the SageMaker AI family that lets busines

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Card #: 13
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon SageMaker Ground Truth?
Back: Amazon SageMaker Ground Truth is a data labeling capability within SageMaker AI that helps create acc

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Card #: 14
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon SageMaker Clarify?
Back: Amazon SageMaker Clarify is a capability that helps detect bias in datasets and models and provides expl

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Card #: 15
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What does Amazon Rekognition do?
Back: Amazon Rekognition is a computer vision service that analyzes images and video to detect objects, faces, t

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Card #: 16
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Beginner
Front: What does Amazon Comprehend do?
Back: Amazon Comprehend is a natural language processing service that analyzes text to detect sentiment, extra

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Card #: 17
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service

Difficulty: Beginner
Front: What does Amazon Textract do?
Back: Amazon Textract extracts printed text, handwriting, and structured data such as form fields and tables fr

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Card #: 18
Domain: 1 - Fundamentals of AI and ML
Category: AWS Service
Difficulty: Intermediate
Front: What does Amazon Forecast do?
Back: Amazon Forecast is an ML-based time-series forecasting capability used to predict future values such as pr

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Card #: 19
Domain: 1 - Fundamentals of AI and ML
Category: Comparison
Difficulty: Intermediate
Front: Supervised learning vs. unsupervised learning: what is the key difference?
Back: Supervised learning uses labeled data with known correct answers, while unsupervised learning uses unlab

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Card #: 20
Domain: 1 - Fundamentals of AI and ML
Category: Comparison
Difficulty: Intermediate
Front: Training set vs. validation set vs. test set: what is each used for?
Back: The training set is used to teach the model. The validation set is used during development to tune the mo

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Card #: 21
Domain: 1 - Fundamentals of AI and ML
Category: Comparison
Difficulty: Intermediate
Front: Classification vs. regression vs. clustering: how do they differ?
Back: Classification predicts a discrete category, such as spam or not spam. Regression predicts a continuous nu

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Card #: 22
Domain: 1 - Fundamentals of AI and ML
Category: Comparison
Difficulty: Intermediate
Front: Real-time inference vs. batch inference: when is each used?
Back: Real-time inference returns predictions immediately for individual requests, ideal for live applications like

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Card #: 23
Domain: 1 - Fundamentals of AI and ML
Category: Comparison
Difficulty: Exam-Ready
Front: Overfitting vs. underfitting: how do they differ in cause and symptom?
Back: Overfitting is caused by a model that is too complex or trained too long, resulting in excellent training per

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Card #: 24
Domain: 1 - Fundamentals of AI and ML
Category: Comparison
Difficulty: Exam-Ready
Front: Accuracy vs. precision vs. recall: what does each metric measure?
Back: Accuracy measures the overall percentage of correct predictions. Precision measures how many predicted p

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Card #: 25
Domain: 1 - Fundamentals of AI and ML
Category: Comparison
Difficulty: Intermediate
Front: AI vs. ML vs. deep learning: how are they related?
Back: AI is the broad field of building systems that simulate intelligent behavior. ML is a subset of AI where sys

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Card #: 26
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: A company has years of labeled historical data showing which loan applications defaulted. Which learning
Back: Supervised learning fits this scenario because labeled outcomes already exist. The model can be trained to

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Card #: 27
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: A company wants to flag unusual transactions in payment data without having any past fraud examples l
Back: Anomaly detection, typically built on unsupervised learning, applies here because there is no labeled fraud

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Card #: 28
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: A retailer wants product recommendations for individual shoppers, while a logistics team wants to forecas
Back: Amazon Personalize fits the recommendation use case because it builds personalized suggestions from user

Card #: 29
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Intermediate
Front: An application needs to automatically detect whether uploaded photos contain inappropriate content befo
Back: This is a computer vision use case, since it involves analyzing the visual content of images. Amazon Rekog

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Card #: 30
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: A data science team needs to label thousands of raw images before training a custom model but has no la
Back: Amazon SageMaker Ground Truth addresses this need by providing data labeling workflows, including acc

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Card #: 31
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "The model performs well on training data but fails on new data." What concept does this de
Back: This phrase is a classic exam clue for overfitting. The model has memorized the training data rather than

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Card #: 32
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "There is no labeled data available." What type of learning does this point to?
Back: This phrase points to unsupervised learning, since supervised learning requires labeled data and unsupervi

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Card #: 33
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "Predict which category an item belongs to." What ML technique does this describe?
Back: This describes classification, which assigns inputs to discrete predefined categories rather than predicting a

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Card #: 34
Domain: 1 - Fundamentals of AI and ML
Category: Scenario

Difficulty: Exam-Ready
Front: Exam clue: “Predict a specific numeric value, such as a price or temperature.” What ML technique does t
Back: This describes regression, which is used to predict continuous numerical outcomes rather than categories o

Card #: 35
Domain: 1 - Fundamentals of AI and ML
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “Use historical purchase data to recommend products to a specific customer.” Which AWS ser
Back: This points to Amazon Personalize, AWS’s managed service for building real-time personalized recommend

Domain 2 - Fundamentals of GenAI

Card #: 36
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Beginner
Front: What is a token in the context of generative AI?
Back: A token is a chunk of text, such as a word or part of a word, that a language model processes as a basic u

Card #: 37
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Beginner
Front: What is a context window?
Back: A context window is the maximum amount of text, measured in tokens, that a model can consider at one

Card #: 38
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Beginner
Front: What is a hallucination in generative AI?
Back: A hallucination is a confidently stated output that is factually incorrect or not grounded in real data. Hall

Card #: 39
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Intermediate
Front: What is temperature in the context of an FM’s inference parameters?
Back: Temperature is an inference parameter that controls the randomness and creativity of a model’s output. L

Card #: 40
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Beginner
Front: What is a foundation model (FM)?
Back: A foundation model is a large, pre-trained AI model that can be adapted for many different tasks without

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Card #: 41
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Beginner
Front: What is a large language model (LLM)?
Back: A large language model is a type of foundation model trained on massive amounts of text to understand a

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Card #: 42
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Intermediate
Front: What is an embedding?
Back: An embedding is a numerical vector representation of text, images, or other data that captures its meanin

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Card #: 43
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Intermediate
Front: What is pre-training?
Back: Pre-training is the initial stage of training a foundation model on a massive, general dataset to learn broa

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Card #: 44
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Intermediate
Front: What is fine-tuning?
Back: Fine-tuning is the process of further training an already pre-trained foundation model on a smaller, task-s

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Card #: 45
Domain: 2 - Fundamentals of GenAI
Category: GenAI Term
Difficulty: Intermediate
Front: What is Reinforcement Learning from Human Feedback (RLHF)?
Back: RLHF is a training technique that incorporates human judgments about output quality into the model's le

Card #: 46
Domain: 2 - Fundamentals of GenAI
Category: AWS Service
Difficulty: Beginner
Front: What is Amazon Bedrock?
Back: Amazon Bedrock is AWS's managed platform for accessing, customizing, and deploying foundation models

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Card #: 47
Domain: 2 - Fundamentals of GenAI
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Titan?
Back: Amazon Titan is a family of foundation models built by AWS and made available through Amazon Bedro

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Card #: 48
Domain: 2 - Fundamentals of GenAI
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Q Business?
Back: Amazon Q Business is an AI-powered assistant designed to help employees find information, answer questi

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Card #: 49
Domain: 2 - Fundamentals of GenAI
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Q Developer?
Back: Amazon Q Developer is an AI coding assistant that helps developers write, troubleshoot, and understand

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Card #: 50
Domain: 2 - Fundamentals of GenAI
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Q in Connect?
Back: Amazon Q in Connect is a generative AI capability for Amazon Connect that gives contact center agents

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Card #: 51
Domain: 2 - Fundamentals of GenAI
Category: AWS Service

Difficulty: Beginner
Front: What is Amazon Lex?
Back: Amazon Lex is a service for building conversational interfaces, such as chatbots and voice bots, powered b

Card #: 52
Domain: 2 - Fundamentals of GenAI
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Kendra?
Back: Amazon Kendra is an intelligent enterprise search service that lets employees search across multiple data s

Card #: 53
Domain: 2 - Fundamentals of GenAI
Category: AWS Service
Difficulty: Intermediate
Front: What is PartyRock?
Back: PartyRock is an Amazon Bedrock playground that lets users build and share simple generative AI apps wi

Card #: 54
Domain: 2 - Fundamentals of GenAI
Category: Comparison
Difficulty: Intermediate
Front: Foundation model (FM) vs. large language model (LLM): how are they different?
Back: A foundation model is the broader category of large, pre-trained models that can be adapted to many task

Card #: 55
Domain: 2 - Fundamentals of GenAI
Category: Comparison
Difficulty: Exam-Ready
Front: Amazon Bedrock vs. Amazon SageMaker AI: how do they differ?
Back: Amazon Bedrock is for accessing and customizing existing foundation models through a managed API wit

Card #: 56
Domain: 2 - Fundamentals of GenAI
Category: Comparison
Difficulty: Intermediate
Front: Amazon Q Business vs. Amazon Q Developer: who is each one for?
Back: Amazon Q Business serves general business users who need help finding information and completing tasks

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Card #: 57
Domain: 2 - Fundamentals of GenAI
Category: Comparison
Difficulty: Intermediate
Front: High temperature vs. low temperature: how do they affect FM output?
Back: A low temperature produces more focused, deterministic, and predictable output, useful for factual tasks.

Card #: 58
Domain: 2 - Fundamentals of GenAI
Category: Comparison
Difficulty: Exam-Ready
Front: Pre-training vs. fine-tuning: how do they differ in purpose and scale?
Back: Pre-training teaches a model broad, general knowledge using a massive dataset and is extremely resource-i

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Card #: 59
Domain: 2 - Fundamentals of GenAI
Category: Comparison
Difficulty: Exam-Ready
Front: Amazon Lex vs. Amazon Q Business vs. Amazon Kendra: what does each one do?
Back: Amazon Lex builds custom conversational chatbot and voice interfaces. Amazon Q Business is an AI assis

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Card #: 60
Domain: 2 - Fundamentals of GenAI
Category: Comparison
Difficulty: Intermediate
Front: Multi-modal models vs. text-only models: what is the key difference?
Back: A multi-modal model can process and generate more than one type of content, such as text, images, and a

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Card #: 61
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Exam-Ready
Front: A company wants to use a pre-built foundation model through an API without managing servers or traini
Back: Amazon Bedrock fits this scenario because it provides managed access to multiple foundation models throu

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Card #: 62
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Intermediate
Front: A development team wants AI assistance writing and debugging code directly inside their IDE. Which ser
Back: Amazon Q Developer helps in this scenario, since it is purpose-built to assist developers with coding, debu

Card #: 63
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Exam-Ready
Front: A chatbot confidently tells a customer about a return policy that does not exist at the company. What G
Back: This illustrates a hallucination, where the model generates plausible-sounding but factually incorrect conte

Card #: 64
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Exam-Ready
Front: An organization wants employees across departments to ask natural-language questions and get answers p
Back: Amazon Q Business fits this scenario, since it is designed to help general business users get answers using

Card #: 65
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Intermediate
Front: A marketing team wants highly creative ad copy ideas, while a legal team wants precise, consistent contra
Back: The marketing team should use a higher temperature to encourage creative and varied output. The legal t

Card #: 66
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "Access pre-built foundation models through a single managed API." Which AWS service doe
Back: This describes Amazon Bedrock, the AWS service built specifically for accessing and customizing foundati

Card #: 67
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "An AI assistant that helps write and troubleshoot code." Which service does this describe?
Back: This describes Amazon Q Developer, which is AWS's AI coding assistant for developers.

Card #: 68
Domain: 2 - Fundamentals of GenAI
Category: Scenario

Difficulty: Exam-Ready
Front: Exam clue: “An employee-facing AI assistant that answers questions using company data.” Which service
Back: This describes Amazon Q Business, which helps employees find information and complete tasks using ente

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Card #: 69
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “Build a custom chatbot dialog flow for a branded application.” Which service does this descr
Back: This describes Amazon Lex, the service used to design custom conversational interfaces such as chatbots a

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Card #: 70
Domain: 2 - Fundamentals of GenAI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “The model confidently states incorrect facts.” What concept does this describe?
Back: This describes a hallucination, a key limitation of generative AI where outputs sound plausible but are no

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Domain 3 - Applications of Foundation Models

Card #: 71
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Beginner
Front: What is zero-shot prompting?
Back: Zero-shot prompting asks a model to complete a task with no examples provided, relying entirely on the m

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Card #: 72
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Beginner
Front: What is one-shot (single-shot) prompting?
Back: One-shot prompting provides exactly one example of the desired input-output pattern before asking the m

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Card #: 73
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is few-shot prompting?
Back: Few-shot prompting provides multiple examples of the desired input-output pattern within the prompt be

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Card #: 74
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is chain-of-thought prompting?
Back: Chain-of-thought prompting guides a model to reason through a problem step by step rather than jumping

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Card #: 75
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is role prompting?
Back: Role prompting instructs a model to respond as if it were a specific persona or expert, such as a financial a

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Card #: 76
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is a system prompt?
Back: A system prompt is a set of instructions provided to a model that establishes its overall behavior, tone, an

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Card #: 77
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is negative prompting?
Back: Negative prompting explicitly tells a model what to avoid in its response, such as a topic, tone, or format.

Card #: 78
Domain: 3 - Applications of Foundation Models
Category: Security and Governance
Difficulty: Exam-Ready
Front: What is prompt injection?
Back: Prompt injection is an attack where malicious instructions are embedded in input data to manipulate a m

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Card #: 79
Domain: 3 - Applications of Foundation Models
Category: AWS Service
Difficulty: Intermediate
Front: What are Amazon Bedrock Knowledge Bases?
Back: Amazon Bedrock Knowledge Bases is a managed RAG capability that connects a foundation model to an

Card #: 80
Domain: 3 - Applications of Foundation Models
Category: AWS Service
Difficulty: Intermediate
Front: What are Amazon Bedrock Agents?
Back: Amazon Bedrock Agents are a capability that lets a foundation model autonomously plan and execute mu

Card #: 81
Domain: 3 - Applications of Foundation Models
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Bedrock Guardrails?
Back: Amazon Bedrock Guardrails is a capability for setting safety limits on what a foundation model can say o

Card #: 82
Domain: 3 - Applications of Foundation Models
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Bedrock Model Evaluation?
Back: Amazon Bedrock Model Evaluation is a managed service for assessing and comparing foundation model pe

Card #: 83
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is fine-tuning in the context of FM customization?
Back: Fine-tuning adapts a pre-trained foundation model to a specific task or domain by continuing training on

Card #: 84
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is continued pre-training?
Back: Continued pre-training updates a foundation model with additional unlabeled domain-specific data withou

Card #: 85
Domain: 3 - Applications of Foundation Models
Category: GenAI Term

Difficulty: Beginner
Front: What is Retrieval Augmented Generation (RAG)?
Back: RAG is a technique that lets a foundation model search and retrieve relevant information from an externa

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Card #: 86
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is a vector database?
Back: A vector database stores data as numerical embeddings and allows fast similarity searches between them.

Card #: 87
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What are embeddings, in the context of RAG?
Back: Embeddings are numerical vector representations of text or other content that capture semantic meaning.

Card #: 88
Domain: 3 - Applications of Foundation Models
Category: GenAI Term
Difficulty: Intermediate
Front: What is chunking?
Back: Chunking is the process of breaking large documents into smaller pieces before generating embeddings and

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Card #: 89
Domain: 3 - Applications of Foundation Models
Category: Comparison
Difficulty: Exam-Ready
Front: RAG vs. fine-tuning: when would you choose one over the other?
Back: Choose RAG when you need the model to use current, frequently changing information without retraining

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Card #: 90
Domain: 3 - Applications of Foundation Models
Category: Comparison
Difficulty: Exam-Ready
Front: Prompt engineering vs. RAG vs. fine-tuning: how do they rank in cost and complexity?
Back: Prompt engineering is the lowest-cost and simplest approach, adjusting only the instructions given to the

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Card #: 91
Domain: 3 - Applications of Foundation Models
Category: Comparison
Difficulty: Exam-Ready
Front: Amazon Bedrock Knowledge Bases vs. Amazon Kendra: how do they differ?
Back: Bedrock Knowledge Bases is purpose-built to ground foundation model responses in company data as part

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Card #: 92
Domain: 3 - Applications of Foundation Models
Category: Comparison
Difficulty: Intermediate
Front: Automated evaluation vs. human evaluation: what are the tradeoffs?
Back: Automated evaluation, using metrics like ROUGE or BLEU, is fast, scalable, and consistent but may miss

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Card #: 93
Domain: 3 - Applications of Foundation Models
Category: Comparison
Difficulty: Intermediate
Front: Zero-shot vs. few-shot prompting: what is the main tradeoff?
Back: Zero-shot prompting requires no examples and is faster to write but may produce less accurate results on c

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Card #: 94
Domain: 3 - Applications of Foundation Models
Category: Comparison
Difficulty: Exam-Ready
Front: Bedrock Agents vs. a traditional API integration: what is the key difference?
Back: A traditional API integration requires a developer to hard-code the exact sequence of calls to make. Bedro

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Card #: 95
Domain: 3 - Applications of Foundation Models
Category: Comparison
Difficulty: Exam-Ready
Front: ROUGE vs. BERTScore: what does each metric evaluate?
Back: ROUGE measures overlap of words and phrases between generated text and a reference, commonly used fo

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Card #: 96
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: A company wants its chatbot to answer questions using a constantly updated product catalog without ret
Back: RAG fits this scenario, since it retrieves current information from the product catalog at query time rathe

Card #: 97
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: A legal team wants a model to consistently write in a very specific contract format and tone unlike typica
Back: Fine-tuning fits this scenario, since the goal is to change the model's underlying behavior and style in a wa

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Card #: 98
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: A company wants to prevent its GenAI application from discussing competitor products or generating offe
Back: Amazon Bedrock Guardrails fits this scenario, since it lets organizations define content filters and topic res

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Card #: 99
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: A team needs a model to answer questions from company documents and autonomously complete a multi-
Back: Amazon Bedrock Knowledge Bases provides grounded answers from company documents, while Amazon B

Card #: 100
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Intermediate
Front: A team wants a fast way to check translation quality across thousands of outputs, but also wants periodic
Back: A combination of automated metrics, such as BLEU, for scalable evaluation and human-in-the-loop evalua

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Card #: 101
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "Connect a foundation model to company documents without retraining." What concept and
Back: This describes RAG, often implemented in AWS through Amazon Bedrock Knowledge Bases, which groun

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Card #: 102
Domain: 3 - Applications of Foundation Models
Category: Scenario

Difficulty: Exam-Ready
Front: Exam clue: “Automatically take actions across multiple systems to complete a task.” Which Bedrock feat
Back: This describes Amazon Bedrock Agents, which let a foundation model plan and execute multi-step actions

Card #: 103
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “Prevent harmful or off-topic content in model responses.” Which Bedrock feature does this d
Back: This describes Amazon Bedrock Guardrails, which enforce content filters and topic restrictions on model o

Card #: 104
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “The model breaks a problem into multiple reasoning steps before answering.” What prompti
Back: This describes chain-of-thought prompting, which guides the model to reason step by step rather than ans

Card #: 105
Domain: 3 - Applications of Foundation Models
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “Evaluate foundation model performance at scale using AWS tooling.” Which service does thi
Back: This describes Amazon Bedrock Model Evaluation, AWS’s managed service for assessing and comparing fo

Domain 4 - Guidelines for Responsible AI

Card #: 106
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Beginner
Front: What is AI bias?
Back: AI bias is a systematic error in a model’s outputs that produces unfair or skewed results, often disadvanta

Card #: 107
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Beginner
Front: What is fairness in responsible AI?
Back: Fairness means an AI system treats all individuals and groups equitably, without systematically disadvant

Card #: 108
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is explainability (XAI)?
Back: Explainability is the ability to describe, in understandable terms, why an AI model produced a specific ou

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Card #: 109
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is interpretability?
Back: Interpretability is the degree to which a human can understand the internal mechanics of how a model arr

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Card #: 110
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is transparency in the context of responsible AI?
Back: Transparency means being open about how an AI system works, what data it was trained on, and its know

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Card #: 111
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: How does hallucination relate to responsible AI?
Back: From a responsible AI lens, hallucinations are a veracity risk, since they produce confidently stated but fal

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Card #: 112
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is toxicity in the context of AI outputs?
Back: Toxicity refers to harmful, offensive, or inappropriate content generated by an AI model. Detecting and fil

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Card #: 113
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is human-in-the-loop evaluation?
Back: Human-in-the-loop evaluation involves people actively reviewing, rating, or correcting AI outputs as part o

Card #: 114
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is a model card?
Back: A model card is structured documentation that describes a model's intended use, performance characterist

Card #: 115
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Beginner
Front: What are the core features of responsible AI?
Back: The core features of responsible AI are bias, fairness, inclusivity, robustness, safety, and veracity. Together

Card #: 116
Domain: 4 - Guidelines for Responsible AI
Category: AWS Service
Difficulty: Intermediate
Front: How does Amazon SageMaker Clarify support responsible AI?
Back: SageMaker Clarify detects potential bias in datasets and model predictions and provides feature importanc

Card #: 117
Domain: 4 - Guidelines for Responsible AI
Category: AWS Service
Difficulty: Intermediate
Front: How does Amazon Augmented AI (A2I) support responsible AI?
Back: Amazon A2I adds human review workflows into ML prediction pipelines, routing low-confidence prediction

Card #: 118
Domain: 4 - Guidelines for Responsible AI
Category: AWS Service
Difficulty: Intermediate
Front: How does Amazon SageMaker Ground Truth support responsible AI?
Back: SageMaker Ground Truth supports responsible AI by helping create accurately labeled, high-quality traini

Card #: 119
Domain: 4 - Guidelines for Responsible AI
Category: AWS Service

Difficulty: Intermediate
Front: How does Amazon Bedrock Guardrails support responsible AI?
Back: From a responsible AI perspective, Bedrock Guardrails enforces content filters, topic restrictions, and grou

Card #: 120
Domain: 4 - Guidelines for Responsible AI
Category: AWS Service
Difficulty: Intermediate
Front: How does Amazon Macie support responsible and safe AI use?
Back: Amazon Macie automatically discovers and classifies sensitive data, such as PII, stored in Amazon S3. In a

Card #: 121
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is SHAP, conceptually, in explainable AI?
Back: SHAP, or SHapley Additive exPlanations, is an explainability technique that estimates how much each inp

Card #: 122
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Intermediate
Front: What is LIME, conceptually, in explainable AI?
Back: LIME, or Local Interpretable Model-agnostic Explanations, is an explainability technique that approximat

Card #: 123
Domain: 4 - Guidelines for Responsible AI
Category: Responsible AI
Difficulty: Beginner
Front: What do AWS's responsible AI principles emphasize?
Back: AWS's responsible AI principles emphasize fairness, explainability, privacy and security, safety, controllabil

Card #: 124
Domain: 4 - Guidelines for Responsible AI
Category: Comparison
Difficulty: Exam-Ready
Front: Historical bias vs. representation bias vs. measurement bias: how do they differ?
Back: Historical bias reflects past societal inequities embedded in the data itself. Representation bias occurs whe

Card #: 125
Domain: 4 - Guidelines for Responsible AI
Category: Comparison
Difficulty: Exam-Ready
Front: Explainability vs. interpretability vs. transparency: how are these related but distinct?
Back: Explainability is the ability to describe why a model made a decision. Interpretability is how understandal

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Card #: 126
Domain: 4 - Guidelines for Responsible AI
Category: Comparison
Difficulty: Intermediate
Front: Human-in-the-loop vs. human-on-the-loop: how do they differ?
Back: Human-in-the-loop means a person actively reviews or approves individual AI decisions before they take ef

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Card #: 127
Domain: 4 - Guidelines for Responsible AI
Category: Comparison
Difficulty: Exam-Ready
Front: SageMaker Clarify vs. Amazon A2I vs. SageMaker Ground Truth: what does each one do?
Back: SageMaker Clarify detects bias and explains model predictions. Amazon A2I routes low-confidence predict

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Card #: 128
Domain: 4 - Guidelines for Responsible AI
Category: Comparison
Difficulty: Intermediate
Front: Fairness vs. accuracy: why can these sometimes conflict?
Back: A highly accurate model overall can still perform poorly or unfairly for a specific subgroup if that group is

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Card #: 129
Domain: 4 - Guidelines for Responsible AI
Category: Comparison
Difficulty: Intermediate
Front: Data bias vs. algorithmic bias: what is the difference?
Back: Data bias originates from the training data itself, such as skewed sampling or historical inequities. Algorith

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Card #: 130
Domain: 4 - Guidelines for Responsible AI
Category: Comparison
Difficulty: Intermediate
Front: Local explanation vs. global explanation: how do they differ?
Back: A local explanation explains why the model made one specific prediction for one specific input. A global e

Card #: 131
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: A hiring model trained on decades of past hiring decisions consistently scores female candidates lower bec
Back: This is historical bias, since the model is learning and reproducing inequities that existed in past human d

Card #: 132
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: A team wants to detect imbalanced demographic representation in their training dataset before training a
Back: Amazon SageMaker Clarify fits this scenario, since it is designed to detect bias in datasets, including imba

Card #: 133
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: A medical diagnosis model produces a prediction with low confidence for an unusual case. What responsib
Back: Human oversight through Amazon A2I should be applied, routing the low-confidence prediction to a quali

Card #: 134
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: A company deploys an AI hiring tool with no documentation of its training data or limitations, and no wa
Back: This violates transparency and explainability, since stakeholders cannot understand the model's data, limi

Card #: 135
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: A customer service chatbot occasionally invents product specifications that do not exist. What risk does t
Back: This represents a hallucination and veracity risk. Amazon Bedrock Guardrails, combined with RAG groun

Card #: 136
Domain: 4 - Guidelines for Responsible AI
Category: Scenario

Difficulty: Exam-Ready
Front: Exam clue: “A hiring algorithm consistently scores one group lower than others.” What concept does this
Back: This describes bias, most likely historical or representation bias, where the model’s outputs systematically

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Card #: 137
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “Why did the model make this decision?” What responsible AI concept does this describe?
Back: This describes explainability, the ability to articulate the reasoning behind a specific model decision.

Card #: 138
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “Detect bias in training data before building a model.” Which AWS tool does this describe?
Back: This describes Amazon SageMaker Clarify, which is built to detect bias in both datasets and trained mode

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Card #: 139
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “A human reviews low-confidence predictions before they are acted on.” Which AWS service
Back: This describes Amazon Augmented AI, or Amazon A2I, which routes low-confidence predictions into huma

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Card #: 140
Domain: 4 - Guidelines for Responsible AI
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: “Workers label training images by hand.” Which AWS service does this describe?
Back: This describes Amazon SageMaker Ground Truth, which supports human-assisted data labeling for trainin

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Domain 5 - Security, Compliance, and Governance for AI Solutions

Card #: 141
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Beginner
Front: What is the AWS Shared Responsibility Model?
Back: The Shared Responsibility Model defines that AWS is responsible for the security of the cloud, meaning th

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Card #: 142
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Beginner
Front: What is the principle of least privilege?
Back: The principle of least privilege means granting a user, role, or service only the minimum permissions neede

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Card #: 143
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Beginner
Front: What is an IAM role?
Back: An IAM role is an identity with a defined set of permissions that can be assumed temporarily by users, ap

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Card #: 144
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Intermediate
Front: What is a VPC endpoint?
Back: A VPC endpoint allows private connectivity between resources in a VPC and supported AWS services, wit

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Card #: 145
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Beginner
Front: What is encryption at rest?
Back: Encryption at rest protects stored data, such as files in Amazon S3 or training datasets, by encrypting it w

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Card #: 146
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Beginner
Front: What is encryption in transit?
Back: Encryption in transit protects data while it moves between systems, such as between an application and an

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Card #: 147
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Intermediate
Front: What is AWS KMS?
Back: AWS Key Management Service, or KMS, is a managed service for creating, storing, and controlling the en

Card #: 148
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Security and Governance
Difficulty: Intermediate
Front: What is data masking?
Back: Data masking is a privacy-enhancing technique that obscures or replaces sensitive data values, such as rep

Card #: 149
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Beginner
Front: What is AWS IAM?
Back: AWS Identity and Access Management, or IAM, is the foundational service for controlling who can access

Card #: 150
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon Macie?
Back: Amazon Macie is an ML-powered data security service that automatically discovers and classifies sensitive

Card #: 151
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Intermediate
Front: What is AWS KMS used for in AI workloads?
Back: AWS KMS manages the encryption keys used to protect AI training data, model artifacts, and other sensi

Card #: 152
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Beginner
Front: What is AWS CloudTrail?
Back: AWS CloudTrail records a complete history of API calls and actions taken within an AWS account, creati

Card #: 153
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service

Difficulty: Intermediate
Front: What is Amazon CloudWatch?
Back: Amazon CloudWatch is a monitoring and observability service that collects metrics, logs, and events to tra

Card #: 154
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Intermediate
Front: What is Amazon SageMaker Model Monitor?
Back: SageMaker Model Monitor continuously tracks a deployed model's performance and data quality, detecting

Card #: 155
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Intermediate
Front: What is AWS Config?
Back: AWS Config continuously records and tracks the configuration state of AWS resources over time, enabling

Card #: 156
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Intermediate
Front: What is AWS Artifact?
Back: AWS Artifact is a self-service portal where customers can access and download AWS's own compliance rep

Card #: 157
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Intermediate
Front: What is AWS Control Tower used for?
Back: AWS Control Tower helps organizations set up and govern a secure, compliant multi-account AWS environ

Card #: 158
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: AWS Service
Difficulty: Intermediate
Front: What does Bedrock logging provide?
Back: Amazon Bedrock logging, often paired with CloudTrail and CloudWatch, captures details of model invocat

Card #: 159
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Comparison
Difficulty: Exam-Ready
Front: AWS CloudTrail vs. Amazon CloudWatch vs. SageMaker Model Monitor: what does each track?
Back: CloudTrail tracks who took what action through AWS API calls, for auditing. CloudWatch tracks operatio

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Card #: 160
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Comparison
Difficulty: Intermediate
Front: SSE-S3 vs. SSE-KMS: how do these S3 encryption options differ?
Back: SSE-S3 uses encryption keys that are fully managed by Amazon S3 with no customer control over the keys

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Card #: 161
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Comparison
Difficulty: Exam-Ready
Front: Shared responsibility for Amazon Bedrock vs. Amazon SageMaker AI: how does the customer's responsibi
Back: With Amazon Bedrock, AWS manages more of the underlying model infrastructure, so the customer is ma

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Card #: 162
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Comparison
Difficulty: Intermediate
Front: Customer managed key vs. AWS managed key: what is the key difference?
Back: A customer managed key is created and controlled by the customer, who can define detailed key policies a

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Card #: 163
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Comparison
Difficulty: Intermediate
Front: Amazon Macie vs. Bedrock Guardrails PII masking: how do their roles differ?
Back: Amazon Macie discovers and classifies sensitive data already stored in Amazon S3. Bedrock Guardrails PI

Card #: 164
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Comparison
Difficulty: Exam-Ready
Front: AWS Config vs. AWS CloudTrail: how do they differ?
Back: AWS Config tracks the state of resource configurations over time and evaluates them against compliance r

Card #: 165
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Comparison
Difficulty: Intermediate
Front: Compliance vs. governance: how are these related but distinct?
Back: Compliance means meeting specific external requirements, laws, or standards, such as passing an audit. G

Card #: 166
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: A company stores its ML training data in Amazon S3 and wants it encrypted. Under the Shared Respons
Back: The customer is responsible for configuring encryption on their own data in S3, even though AWS provide

Card #: 167
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: A healthcare company wants to ensure traffic to Amazon Bedrock never crosses the public internet. What
Back: They should use a VPC endpoint powered by AWS PrivateLink, which allows private connectivity to Bedr

Card #: 168
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: A compliance team needs a complete record of every API call made to Amazon Bedrock across the organi
Back: AWS CloudTrail provides this record, since it logs all API activity, including calls made to Amazon Bedro

Card #: 169
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: A security team suspects an S3 bucket used for AI training data may contain personally identifiable infor
Back: Amazon Macie should be used, since it automatically discovers and classifies sensitive data, including PII,

Card #: 170
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario

Difficulty: Intermediate
Front: An organization wants to formally track AI model performance, compliance evidence, and configuration c
Back: A combination of SageMaker Model Monitor for performance, AWS Config for configuration tracking, and

Card #: 171
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "Determine who called which AWS API and when." Which AWS service does this describe?
Back: This describes AWS CloudTrail, which logs account activity and API calls for auditing purposes.

Card #: 172
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "Detect personally identifiable information stored in an S3 bucket." Which AWS service does
Back: This describes Amazon Macie, which automatically discovers and classifies sensitive data like PII in Amaz

Card #: 173
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "A deployed model's performance is degrading over time." Which AWS service does this descr
Back: This describes Amazon SageMaker Model Monitor, which tracks deployed model performance and detects

Card #: 174
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "Prevent Bedrock traffic from using the public internet." Which AWS feature does this descri
Back: This describes a VPC endpoint using AWS PrivateLink, which provides private connectivity to Bedrock w

Card #: 175
Domain: 5 - Security, Compliance, and Governance for AI Solutions
Category: Scenario
Difficulty: Exam-Ready
Front: Exam clue: "Download AWS compliance reports for regulators." Which AWS service does this describe?
Back: This describes AWS Artifact, the self-service portal for accessing AWS's own compliance reports and certi

End of Flashcard Deck - 175 cards total - Task 11 of 15